



# The Business Value of Amazon Web Services: Asia/Pacific

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## BUSINESS VALUE HIGHLIGHTS



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**318%**  
5-year ROI

**12 months**  
to payback

**45%**  
lower 5-year cost  
of operations

**25%**  
lower cost of  
infrastructure

**58%**  
more IT staff time  
for innovation

**49%**  
fewer unplanned outages

**81%**  
less number of hours per  
deployment

**26%**  
higher developer productivity

**\$22.0M**  
higher revenue enabled per  
year per organization

## Executive Summary

In today's digital-first world, the cloud plays a critical role in helping organizations gain faster access to new technologies in mission-critical areas. These cloud services cover needs across a diverse technology ecosystem, including compute services, data services, application framework services, and usage multiplier services. Organizations, developers, and IT organizations are relying upon these cloud services to drive innovation, automation, and insights of value. Furthermore, cloud continues to be defined by fundamental cloud governance, consisting of the managing, optimizing, migrating, and securing of diverse cloud resources and data sets. Increasingly, cloud selection processes are going beyond technology considerations to look at how IT enables business outcomes. IDC expects that by 2023, 40% of the world's 2,000 largest public companies will reset cloud selection processes to focus on business outcomes rather than IT requirements, valuing access to service providers' portfolios. Technologies such as data lakes combined with artificial intelligence (AI) and machine learning (ML) are helping companies deliver value to their customers, resulting in increased revenue as demonstrated in this study. Organizational leaders are increasingly expecting cloud providers to partner with them as they migrate on-premises estates to the cloud and make the transition to cloud-native development and governance, including both cost governance and IT governance. CIOs and CTOs will require cloud providers to help them achieve business case goals directly related to corporate strategy. Business value studies are an important part of that objective.

IDC conducted in-depth interviews of 15 organizations using various Amazon Web Services (AWS) solutions in the Asia/Pacific (APAC) region. These APAC-based customers reported achieving strong value by leveraging AWS to provide an agile, scalable, and high-performing IT platform for their businesses while gaining benefits from IT cost and staff efficiencies. These organizations operate in diverse markets, characterized by different cost and competitive pressures.

**IDC calculates that interviewed APAC organizations will realize benefits worth an average five-year ROI of 318% by:**

- ▶ **Optimizing IT infrastructure costs** associated with delivering compute, storage, and other IT resources to their businesses through flexibility in provisioning and enabling the use of new technologies that allow for more cost-effective use of IT resources
- ▶ **Empowering IT teams** to spend less time on day-to-day activities and thereby refocus on innovative and business-enabling activities
- ▶ **Lowering overall cost of operations** for running equivalent workloads by reducing IT infrastructure spend and staff time requirements for day-to-day activities
- ▶ **Establishing more effective development teams** by reducing the time to market and allowing for more frequent delivery of new applications, features, and updates
- ▶ **Reducing operational risk and related costs** by minimizing the frequency and duration of unplanned outages that affect the user experience and business outcomes
- ▶ **Realizing better business results** by addressing and winning more new business opportunities and better maintaining existing customers with higher-quality services and products

## Situation Overview

Using lessons learned from ecommerce where fluctuating demand for computing resources is a given, Amazon kicked off the Amazon Web Services initiative in 2006. Service-oriented architecture (SOA) gained popularity in the mid-1990s to reduce the complexity of software by breaking them up into components delivered as services and connected through common standards. The SOA experience that predated the AWS launch gave Amazon an advantage in building a distributed services architecture offering that fits very well with end-user needs. Using customer feedback, the services offered by AWS have expanded from the initial portfolio that delivered base infrastructure services to higher-end services that are all based on a highly available infrastructure delivered in an abstracted and automated fashion. AWS also provides customers with multiple migration tools to assist them to move compute and database workloads to the public cloud. The success of Amazon is demonstrated by Amazon moving from number 29 on the Fortune 500 list in 2015 to number 2 in 2021.

## Business Challenges

The combination of lower-cost cloud-delivered services and agility from open source and cloud-native application development tools is leading to lower infrastructure and labor costs for organizations. In addition, cloud-delivered services enable companies with a faster ability to serve customer needs. As compared with on-premises IT models, those using cloud-provisioned services are much more quickly able to respond to business needs. Even for planned IT and software development initiatives, development teams for new projects no longer need to wait weeks or months for new resources to be provisioned. As a result, businesses that fail to take advantage of technological innovation with the cloud often struggle to maintain their competitive advantage. One of the AWS customers interviewed in this study, a manufacturer, summed it up well in noting

that its strategy to grow and diversify via acquisitions is enabled with a cloud platform because it allows the manufacturer to quickly expand IT platforms and governance and, at the same time, not be limited by supply chain risks of getting on-premises hardware.

To surpass competition, today's C-suite has to lead digital transformation, including net-new application development, along with modernization of legacy application estates. This is resulting in impacts at all levels of business and IT strategy. Cloud providers and the wider cloud partner ecosystem are stepping up to serve strategic partnership roles with their customers.

## Key Cloud Revenue Growth Trends

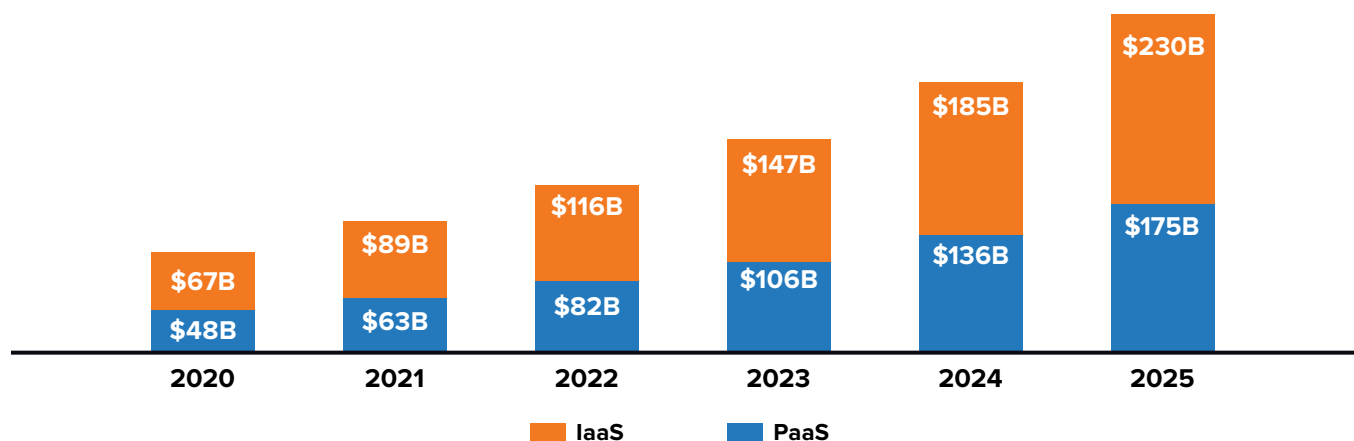
**Figure 1** shows the worldwide cloud vendor revenue forecast in infrastructure as a service (IaaS) and platform as a service (PaaS) for 2020–2025. Together, IaaS and PaaS are projected to grow at a healthy 2020–2025 compound annual growth rate (CAGR) of 29%, from just over \$100 billion in 2020 to reach \$400 billion in 2025. The continued growth in revenue in the cloud market illustrates cloud adoption and speaks to the value that organizations place on the agility gained by developing and deploying applications via cloud services.

In 2020, AWS held 46% of the IaaS market, growing at 32% year over year, and held 15% of the PaaS market, growing at 30% year over year. These market shares and growth rates indicate the success customers are finding with AWS (see *Worldwide Public Cloud Infrastructure as a Service Market Shares, 2020: Data Growth, Edge Use Cases, and Hybrid Deployment Take Center Stage*, IDC #US47350821, July 2021, and *Worldwide Competitive Public Cloud Platform as a Service Market Shares, 2020: The Demand for Application Development Leveraging PaaS Continues Strong Across Vendors as DX Accelerates*, IDC #US48033221, July 2021).

**FIGURE 1**

### Worldwide Whole Cloud Revenue by Consumption Model, 2020–2025

(\$B per year)



Note: For more details, see *Worldwide Whole Cloud Forecast, 2021–2025: The Path Ahead for Cloud in a Digital-First World* (IDC #US47397521, September 2021).

Source: IDC, 2021

# Amazon Web Services

AWS emerged from Amazon's development of a highly scalable compute and storage infrastructure to enable the vendor's large-scale ecommerce business. These compute and storage technologies were the backbone of AWS' early product offerings, including Amazon Elastic Compute Cloud (Amazon EC2) and Amazon Simple Storage Service (Amazon S3), both of which served to abstract the infrastructure for users. The availability of cloud-delivered services led to rapid adoption by start-ups, which found the platform to be flexible, agile, extensible, and available at a low cost, especially compared with managing one's own datacenter infrastructure.

Responding to continuously evolving customer needs, Amazon built a wide array of higher-level services, leading to increased adoption by enterprises that found value in application modernization and the benefits of decreased costs, increased productivity, improved resilience, and agility. Today, AWS' portfolio of services covers analytics, business applications, blockchain, security/identity/compliance, storage, database, development tools, compute, media services, hybrid architecture, Internet of Things, and machine learning (see **Figure 2**).

**FIGURE 2**  
**Overview of AWS Services**

## ANALYTICS

- Analytics
- Data Exchange
- Data Lake
- Data Pipelines
- Data Warehouse
- Elasticsearch
- Streaming
- ETL
- Hadoop/Spark
- Interactive SQL Queries
- Visualizations

## BUSINESS APPLICATIONS

- Contact Center
- Sharing & Collaboration
- Online Meetings & Chat
- Voice-Enabled Workplace
- Unified Communications
- Mobile & Web Apps Without Programming

## BLOCKCHAIN

- Blockchain Templates
- Ledger Database
- Managed Blockchain

## SECURITY, IDENTITY & COMPLIANCE

- Access Control
- Assessment & Reporting
- Configuration Compliance
- Data Protection
- DDoS Protection Identity Management
- Key Management & Storage
- Monitoring & Logging
- Resource Management
- Threat Detection
- Web Applications Firewall

## STORAGE

- Archive Storage
- Backup and Restore
- Block Storage
- Data Transfer
- Edge Processing and Computing
- File Storage
- High-Performance File System
- Hybrid Cloud Storage
- Object Storage
- Windows File System

## DATABASE

- Relationship Database
- High-Performance Relational Database
- Built for the Cloud
- Managed MariaDB
- Managed MySQL
- Managed Oracle
- Managed PostgreSQL
- Managed SQL Server
- Purpose-Built Database
- Document Database
- Graph Database
- In-Memory Caching
- Key-Value Store Database
- Ledger Database
- Time Series Database

## DEVELOPMENT TOOLS

- Analyze & Debug
- Application Life-Cycle Management
- Authoring
- Build & Test
- Containers
- DevOps Resource Management
- One-Click App Development
- Patching
- Pipeline Orchestration
- Resource Templates
- Triggers

## COMPUTE

- Compute
- Auto Scaling
- Batch Jobs
- Event-Driven Serverless Computing
- Instance Types
- Managed Virtual Private Servers
- Managed Repository for Serverless Apps
- Run & Manage Web Apps
- Serverless Compute
- Virtual Servers
- Containers
- Container Service
- Managed Kubernetes
- Store & Retrieve Docker Images

## MEDIA SERVICES

- Live Video Transport
- Media Storage
- Transcoding
- Video Origination and Packaging
- Video Personalization and Monetization
- Video Processing & Delivery
- Video Streaming Analysis

## HYBRID ARCHITECTURE

- AWS Services On-Premises
- Data Integration
- Integrated Devices & Edge Systems
- Integrated Identity & Access
- Integrated Networking
- Integrated Resource & Deployment Management
- VMWare Cloud on AWS
- Integrated 5G

## INTERNET OF THINGS (IoT)

- Rules Engine
- Device Analytics
- Device SDK
- Device Gateway
- Device Shadows
- Event Detection & Response
- Local Compute
- Local Data Collection
- Management & Security
- Microcontroller Operating System
- Visual Applications Development
- Registry

## MACHINE LEARNING

- ML Frameworks
- Deep Learning AMLs & Containers
- Hardware Acceleration
- ML at the Edge
- Tensorflow, Pytorch, Mxnet
- Sagemaker
- Automatic Model Tuning
- Data Labeling
- Hosted Notebooks
- ML Marketplace
- Model Hosting
- Model Optimization
- Model Training
- Pre-Built Algorithms
- Topic Modeling
- Deep Learning Models
- Reinforcement Learning
- Spot Instances
- Batch Predictions
- Real-Time Predictions
- AI Services
- Chatbots
- Entity Extraction
- Face Analytics
- Face Search
- Forecasting
- Image Labeling
- Nature Language Processing
- Personalization & Recommendation
- Sentiment Analysis
- Speech Transcription
- Text & Data Extraction
- Text to Speech
- Translation
- Video & Image Analysis
- Content Moderation

Source: AWS, 2022

# The Business Value of Amazon Web Services

## Study Demographics and Methodology

IDC conducted research exploring the value and benefits for organizations using AWS based in the Asia/Pacific region. The project included 15 in-depth interviews with organizations that have accrued significant practical knowledge of the benefits and costs related to use of AWS.

**The purpose of the study was to develop an in-depth, nontheoretical understanding of the impact of AWS on study participants' IT costs, capabilities, and business results covering topics that included the following:**

- ▶ The comparative cost of providing IT resources, including compute, storage, and other IT resources
- ▶ IT staff capabilities and time required to perform day-to-day activities
- ▶ Time required to provide new compute, storage, and other IT capacity to development and business teams as required
- ▶ Development team capabilities and productivity levels
- ▶ Frequency and impact of unplanned outages and performance degradation on employee productivity and business results
- ▶ Business results, including the ability to compete for, win, and maintain customers

For further detail on IDC's Business Value Study approach, see **Appendix A: Methodology** and **Appendix B: Quantified Benefits of Use of AWS**.

**Table 1** (next page) presents study firmographics for the interviewed AWS customers. Collectively, interviewed APAC-based organizations had the profile of a large enterprise with an average of 14,353 employees (median of 10,000 employees), with annual revenue averaging \$3.7 billion (median of \$3 billion). Seven geographical locations in the APAC region were represented: Australia, Hong Kong, India, Japan, New Zealand, the Philippines, and Singapore. Study participants offered experiences from a range of industry verticals, including, financial services, food and beverage, healthcare, insurance, manufacturing, marketing, market research, media and entertainment, pharmaceuticals, professional services, retail, telecommunications, and utilities.

**TABLE 1**  
**Demographics of Interviewed Organizations**

|                                 | Average                                                                                                                                                                                                                     | Median |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Number of employees             | 14,353                                                                                                                                                                                                                      | 10,000 |
| Number of IT staff              | 394                                                                                                                                                                                                                         | 375    |
| Number of business applications | 145                                                                                                                                                                                                                         | 80     |
| Annual revenue/budget           | \$3.7B                                                                                                                                                                                                                      | \$3.0B |
| Countries                       | Australia (2), Hong Kong, India (5), Japan (2), New Zealand, the Philippines (2), and Singapore (2)                                                                                                                         |        |
| Industries                      | Financial services (2), telecommunications (2), food and beverage, healthcare, insurance, manufacturing, marketing, market research, media and entertainment, pharmaceuticals, professional services, retail, and utilities |        |

n = 15; Source: IDC in-depth interviews, February 2022

## Choice of Amazon Web Services

As reliance upon cloud-based solutions continues to grow, the question for more organizations is not *whether* to use cloud-based services, but rather *which* cloud providers to select. Competition among cloud-based providers is significant, and organizations do not lack choices when selecting a provider.

The APAC organizations interviewed for this study all selected AWS over competing cloud solutions as they migrated or deployed more business applications to the cloud. When queried about the reasoning behind their selection of AWS, responses coalesced around the desire for a platform that offered a combination of innovation, optimization, and speed. Study participants commonly noted scalability, availability, and reliability as key reasons for choosing AWS. Access to compute and other IT resources on demand and the ability to negate the need to build and support internal infrastructure were also reported as key to AWS selection, as was the broad range of service offerings provided by AWS.

### Interviewed AWS customers described these selection criteria in their own words:

- ▶ **Ability to experiment while still running the core business (data scientist, telco, Philippines):**  
*“One of the key things we use AWS for is experimentation. For example, our real-time campaign system was easily deployed, and we have scalability and high availability on some critical revenue-generating applications that are running 24 x 7 without any downtime.”*
- ▶ **Flexibility and ability to drive innovation (cloud platform head, utilities, New Zealand):**  
*“Computing on demand is the number 1 benefit with AWS, as well as access to a wide variety of services and the ability for us to quickly spin up innovative projects without having to deal with hardware.”*

► **Improved management, more resilient/redundant (enterprise systems architect, professional services, Japan):**

*“One of the real differences in our IT strategy is to use and consume PaaS rather than building infrastructure from scratch. For example, two years ago, we switched from our own managed MySQL database to RDS to get rid of the headache of patching and running our own clusters, as well as to have a fully resilient and redundant database as a service.”*

► **Faster infrastructure provisioning that enables development and optimizes related costs, (head of wholesale banking technology, financial services, India):**

*“With AWS, we can now take very quick decisions on infrastructure — decisions that previously took weeks now can be done in a few hours or days. This has sped up our releases and reduces resource cost proportionally.”*

## Use of Amazon Web Services by Interviewed Organizations

APAC organizations participating in this study reported significant organization-wide reliance upon the services offered by AWS. Most have used AWS for at least two years, and they tied more than half (55% on average) of their revenue directly to applications and services that run on AWS-provided platforms. The scale of study participants' use of AWS is reflected in their use of core services such as EC2, with interviewed organizations using an average of 484 EC2 instances and an average maximum of 1,621 EC2 instances. The substantial range that exists between the average number of EC2 instances and the maximum number of EC2 instances demonstrates the dynamic scalability afforded by AWS (see **Table 2**).

**TABLE 2**  
**AWS Use by Interviewed Organizations**

|                                     | Average | Median |
|-------------------------------------|---------|--------|
| Average number of AWS EC2 instances | 484     | 100    |
| Maximum number of AWS EC2 instances | 1,621   | 175    |
| Number of databases                 | 26      | 20     |
| Number of terabytes of storage      | 42,632  | 127    |
| Number of applications              | 85      | 16     |
| Number of users of applications     | 2,276   | 1,475  |
| Revenue supported                   | 55%     | 50%    |

n = 15; Source: IDC in-depth interviews, February 2022

## Business Value and Quantified Benefits of Amazon Web Services

Interviewed APAC organizations reported that they have leveraged various AWS solutions to establish cost-effective and cost-efficient IT platforms. They linked their use of AWS to much enhanced agility and scalability, which enables them to serve customers in a more timely and robust fashion and deliver improved digital experiences to both customers and employees.

### They provided specific examples of these benefits:

- ▶ **Licensing and infrastructure operational cost savings (regional head of APAC, manufacturing, Singapore):**  
*"We have definitely saved on licensing costs with AWS, as well as facility and power and cooling costs over the last two years. We've saved at least US\$2 million this time."*
- ▶ **Scalable performance for data-related activities (vice president of technology, media and entertainment, India):**  
*"The first and the best advantage of AWS is that it gives high scale to our business. In our industry, a lot of content analysis needs to be done, and it used to require a lot of manual activity. Now with AWS, we get data analyses and the metadata, and it goes very fast with their recommendations and content analysis engine."*
- ▶ **Best platform for meeting business needs (data scientist, telco, Philippines):**  
*"We moved to AWS to be able to quickly deploy new applications and to satisfy the needs of the business more quickly. ... In the past, we would just lift and shift applications, but now, we are re-architecting them to be more cloud-native with AWS."*

**IDC estimates that interviewed APAC AWS customers will achieve benefits worth an annual average of \$21,300 per AWS EC2 virtual instance (\$10.3 million per organization) in the following areas (see Figure 3, next page):**

- ▶ **IT staff productivity gains:**  
 Study participants have reduced the day-to-day burden on IT infrastructure and help desk teams and leveraged enhanced functionality and agility to empower their security and development teams. For the purposes of this analysis, IDC classified development team productivity gains as IT staff-related benefits, although improved development throughput and timeliness also clearly support better business results. As a result, these teams work more efficiently and productively, with IDC projecting that interviewed APAC-based AWS customers will achieve staff efficiencies and productivity gains worth an annual average of \$13,200 per AWS EC2 virtual instance (\$6.4 million per organization).
- ▶ **Business productivity benefits:**  
 Study participants have increased their revenue by winning more business opportunities and losing fewer existing customers by improving the quality, timeliness, and security of their services and products. IDC calculates the value of higher net revenue at an annual average of \$5,900 per AWS EC2 virtual instance (\$2.9 million per organization).

► **Risk mitigation benefits — user productivity gains:**

Study participants have reduced the frequency and impact of unplanned outages on business operations and enhanced their ability to ensure compliance with regulatory requirements. IDC values resultant productivity and net revenue gains at an annual average of \$1,100 per AWS EC2 virtual instance (\$551,100 per organization).

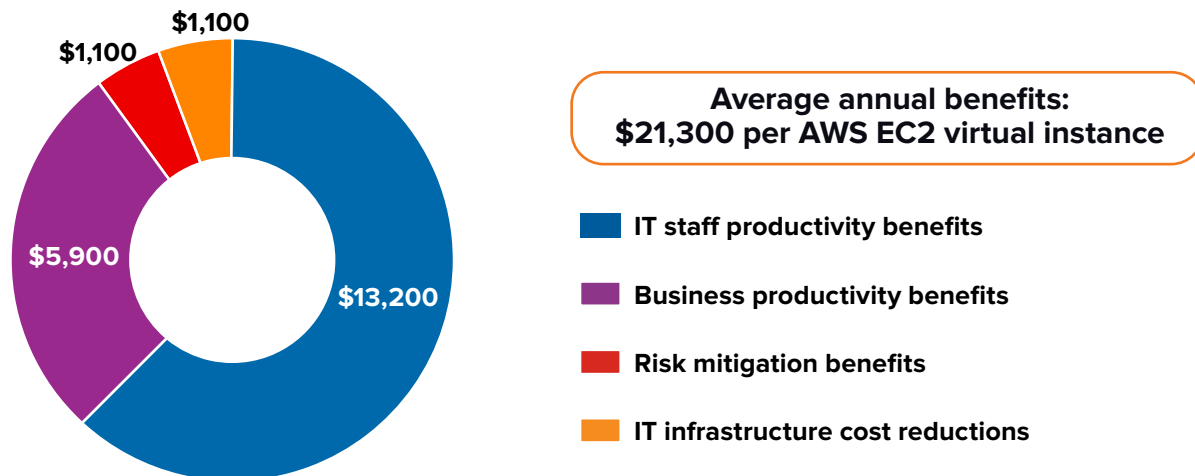
► **IT infrastructure cost reductions:**

Study participants have established more cost-effective IT foundations for their organizations by better tying IT infrastructure resource use to actual business demand. IDC estimates that they will save an annual average of \$1,100 per AWS EC2 virtual instance (\$521,600 per organization).

**FIGURE 3**

### Average Annual Benefits per EC2 Virtual Instance

(\$ per EC2 virtual instance)



n = 15; Source: IDC in-depth interviews, February 2022

## Optimizing Spending on IT Infrastructure Resources

Interviewed organizations reported realizing significant operational efficiency gains with AWS. In comparison with on-premises environments, AWS has provided them a more cost-effective platform for delivering IT resources in support of business operations.

### Study participants cited key AWS functionalities and features that generate these cost efficiencies, including:

► **Enhanced scalability:**

AWS allows organizations to scale their use of IT resources to accommodate evolving business needs, whether that means scaling up or scaling down. As a result, spend reflects actual business needs. This compares favorably with on-premises IT infrastructures, which often do not scale as easily, and enables significant cost savings through avoiding overprovisioning while not throttling business growth.

► **Dynamic flexibility:**

AWS affords organizations the ability to experiment with and adopt new technologies such as containerization and microservices that can provide more streamlined and cost-effective IT foundations for their businesses.

► **Increased visibility:**

AWS provides better visibility into actual usage of resources in comparison with on-premises environments, which allows study participants to better monitor and adjust actual use and costs.

► **Licensing efficiencies:**

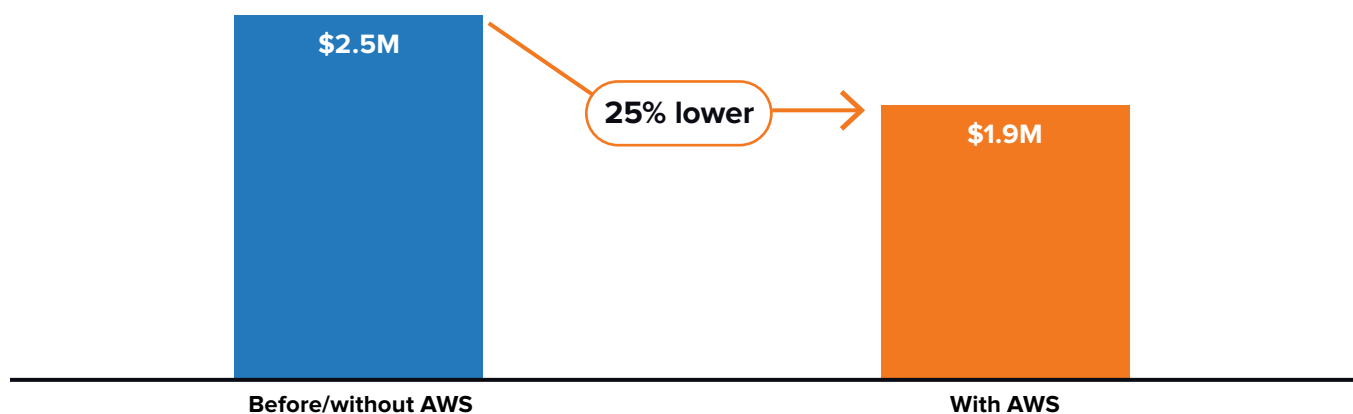
AWS gives interviewed organizations the ability to consolidate and optimize licensing costs and limit staff overhead required to manage and monitor license use.

These benefits combine to provide substantial operational cost savings. Interviewed APAC customers provided real-world examples of applying these capabilities to their operational and cost benefit. For example, the data science lead at a Hong Kong-based insurance company explained: “Using AWS is more efficient because we can scale up and scale down based on the usage, while the stuff in-house we pay for no matter what’s the volume — we save 50% compared with on-premises.”

Overall, AWS has enabled study participants to substantially reduce direct IT infrastructure costs. In sum, they reported average annualized cost reductions of 25%, with annualized IT infrastructure cost savings of over \$600,000 per organization (see **Figure 4**).

**FIGURE 4**  
**Annualized Infrastructure Costs**

(\$ per organization per year)



n = 15; Source: IDC in-depth interviews, February 2022

## Empowering IT Teams

Study participants have not only realized substantial savings in IT infrastructure costs but also achieved significant IT team efficiency gains. They reported that AWS functionality allows IT team members to spend far less time on day-to-day management and support (i.e., keeping-the-lights-on activities). Interviewed AWS customers from the APAC region attributed increased efficiency of IT teams to the features, functionalities, and support provided by AWS; these all translated to less time spent on routine management and support tasks. This has resulted in a substantial shift in focus for their core IT infrastructure teams, freeing up much more time for innovative work.

IT team efficiencies directly benefit all study participants by optimizing total costs for running equivalent workloads. However, these efficiencies also have more indirect but equally important impacts by opening up bandwidth for IT infrastructure and other teams to focus on innovation and capturing value from new technologies. Ultimately, this shift in focus plays a significant role in how well their overall IT operations support business operations, especially as their lines of business increasingly require IT to provide a foundation for technological and competitive differentiation.

### Interviewed AWS customers from the APAC region provided specific examples of these types of gains:

► **Improved access to new technologies and driving digital transformation (vice president of technology, media and entertainment, India):**

*“With AWS, we have a lot of opportunities for learning about new technologies. We can do evaluations and try out projects because AWS is scalable and flexible. We can focus more on the operations side and manage it, rather than getting our hands into installation and configuration. ... As a result, we are able to better manage our environment and spend more time on our overall digital transformation.”*

► **Faster to market due to built-in functionalities (data scientist, telco, Philippines):**

*“With AWS, we’re able to quickly deliver a specific deployment of an application or set up the infrastructure, so we’re able to do more. For example, instead of just deploying an application every three months, we can deploy applications in one week because of the automations we implement using some of the cloud components because they already have their built-in scaling or auto-healing.”*

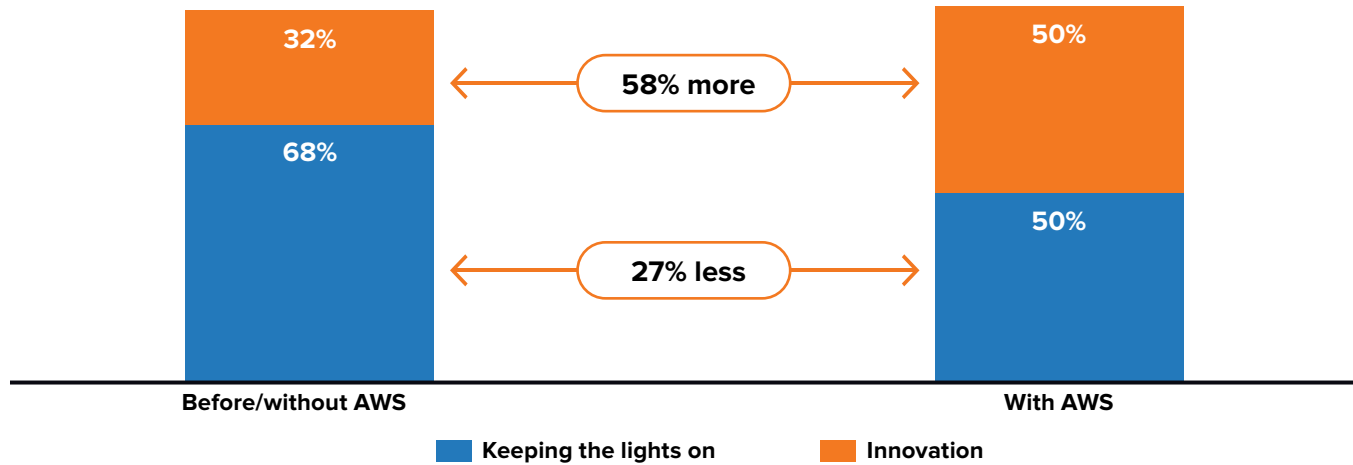
► **Day-to-day efficiencies changing IT teams’ focus (head of wholesale banking technology, financial services, India):**

*“Our IT infrastructure group is better organized with AWS and able to work on a variety of tasks and develop additional skills. We spend much less time putting out fires or dealing with scalability issues. ... The automation and the ease of provisioning and scaling have allowed people to better manage their overall responsibilities.”*

Study participants reported that AWS has enabled their IT infrastructure teams to spend 58% more time on innovative work and 27% less time on routine tasks, representing a significant shift toward innovation and forward-looking work (see **Figure 5**, next page).

**FIGURE 5**  
**Impact on IT Infrastructure and Administrative Team Activities**

(% of time)



n = 15; Source: IDC in-depth interviews, February 2022

IT efficiencies enabled by AWS generate significant value for study participants. Interviewed APAC customers reported requiring 38% less staff time to manage and administer infrastructure with AWS for equivalent workloads and applications, which equates to saving more than 30 FTEs per organization. This represents not only a substantial operational cost efficiency but also a shift in staff utilization patterns (i.e., away from day-to-day management activities to more innovative and profitable activities and creating opportunities to better leverage the skills, talent, and capabilities of these teams to drive improved business results) (see **Table 3**). Study participants also reported achieving noticeable efficiencies for their help desk (34%), database administrator (22%), and security teams (29%) (refer to **Table 7** for details).

**TABLE 3**  
**Impact on IT Infrastructure and Administration Teams**

|                                                   | Before/<br>Without AWS | With AWS | Difference | Benefit |
|---------------------------------------------------|------------------------|----------|------------|---------|
| Equivalent FTEs required for same workloads       | 84                     | 52       | 32         | 38%     |
| Staff hours per AWS EC2 virtual instance per year | 325                    | 200      | 125        | 38%     |
| Value of equivalent FTE time required             | \$8.4M                 | \$5.1M   | \$3.2M     | 38%     |

n = 15; Source: IDC in-depth interviews, February 2022

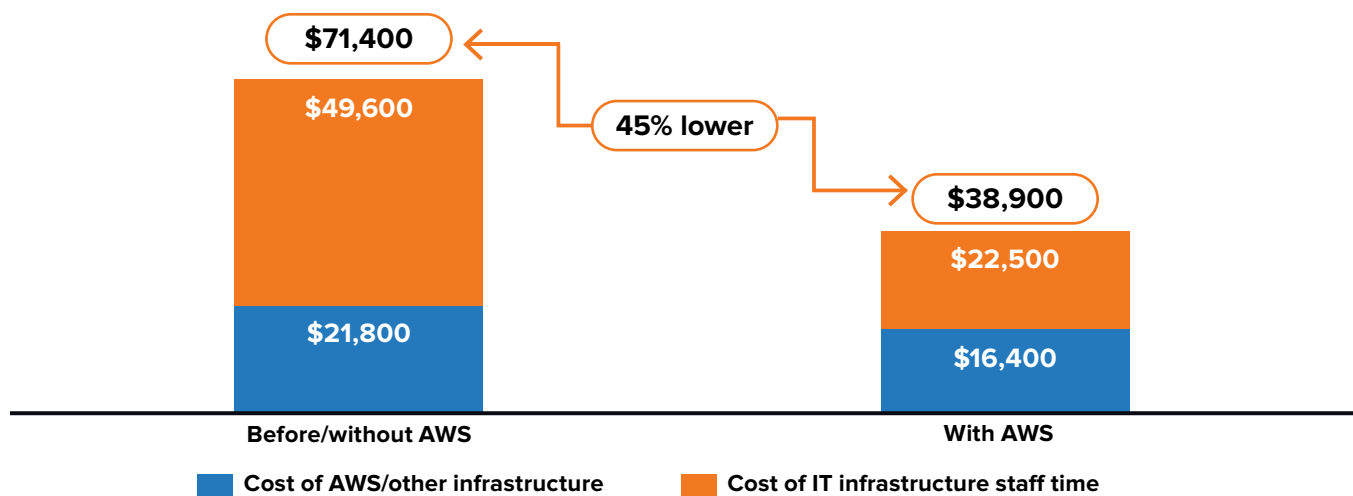
## Lowering Cost of Operations

Staff efficiencies and lower operational costs, as previously detailed, combine to make AWS a notably more cost-effective IT platform for APAC organizations participating in this study. They reported lowering the combined costs of IT resources and staff time devoted to running and administering IT infrastructures for equivalent workloads by an average of 45%. Thus IDC analysis shows that they will achieve average savings of more than \$32,000 per AWS EC2 virtual instance over five years (\$15.7 million in savings per organization over five years) (see **Figure 6**).

**FIGURE 6**

### Five-Year Cost of Operations per EC2 Virtual Instance

(\$ per EC2 virtual instance)



n = 15; Source: IDC in-depth interviews, February 2022

## Improving the Capabilities of Development Teams

Study participants reported that AWS has provided them access to IT infrastructure resources in a way that benefits their agility, scalability, and flexibility. As previously noted, these benefits contribute to interviewed APAC organizations' ability to establish and maintain a cost-effective IT infrastructure. As importantly, enhanced agility drives deeper and substantial value for most organizations from a business operations perspective, including enabling their development activities, extending their business operations more readily, and accessing cost-effective resources on demand to experiment or innovate.

## Interviewed AWS customers provided specific examples of the value of greater agility and flexibility:

### ► Elastic scalability with Lambda and other services of AWS (head of IT infrastructure, financial services, Singapore):

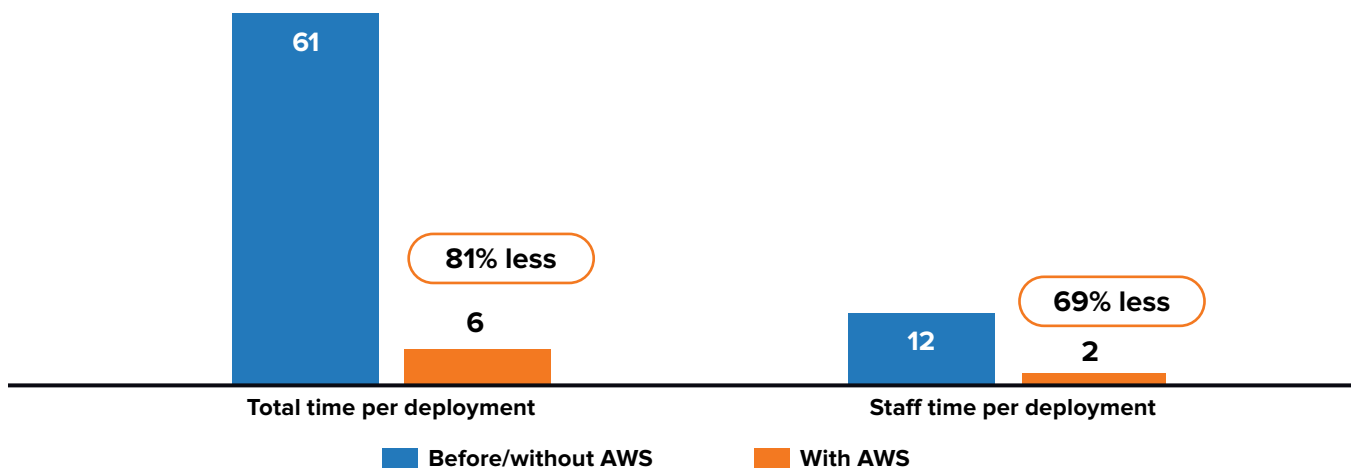
*"AWS provides us with elastic flexibility. We set up our AWS environment in one week, which normally would take three to four months in our datacenter. ... We have used Lambda to automate some processes, and it was really useful because we don't have to deploy servers."*

### ► Ease of scaling with Lambda to enable application of AI and other new technologies (data science lead, insurance, Hong Kong):

*"Our in-house AI research team needed to run an experiment using a large amount of data. If we were doing it on premises, we wouldn't have enough resources to do a big experiment. With AWS, we can just scale up, and that makes us move a lot faster. Sometimes we run very high CPU-intensive tasks, and that's when AWS Lambda will be used because we can scale up horizontally using this serverless interface."*

**Figure 7** quantifies the agility-related benefits in terms of the time required to deploy the compute/storage resources needed by the development and business teams of the organizations participating in the study. On average, the staff time required to deploy new compute/storage resources was reduced by 69%, while the total time required to deploy compute/storage resources was slashed by 81%.

**FIGURE 7**  
**Impact on IT Agility**  
(# of hours per deployment)



n = 15; Source: IDC in-depth interviews, February 2022

Study participants linked enhanced IT agility and flexibility with AWS to a significant positive impact on their development teams' capabilities and productivity levels. With AWS, development teams deliver new software functionality and features more frequently and faster than on their legacy on-premises architectures. They can also make changes and updates to existing applications more readily, speeding up those activities by an average of 23%.

### Interviewed AWS customers in APAC provided details about the positive impact on their development teams' capabilities:

► **Agile development because of the ability to pay only for what is used (enterprise systems architect, professional services, Japan):**

*"AWS has enabled us from a programming perspective because we can spin up extra development environments for one-half day and do load balancing or some performance tests. When we're finished, we take it all down again, and we're charged for just that use. We could never have done that in a physical environment before."*

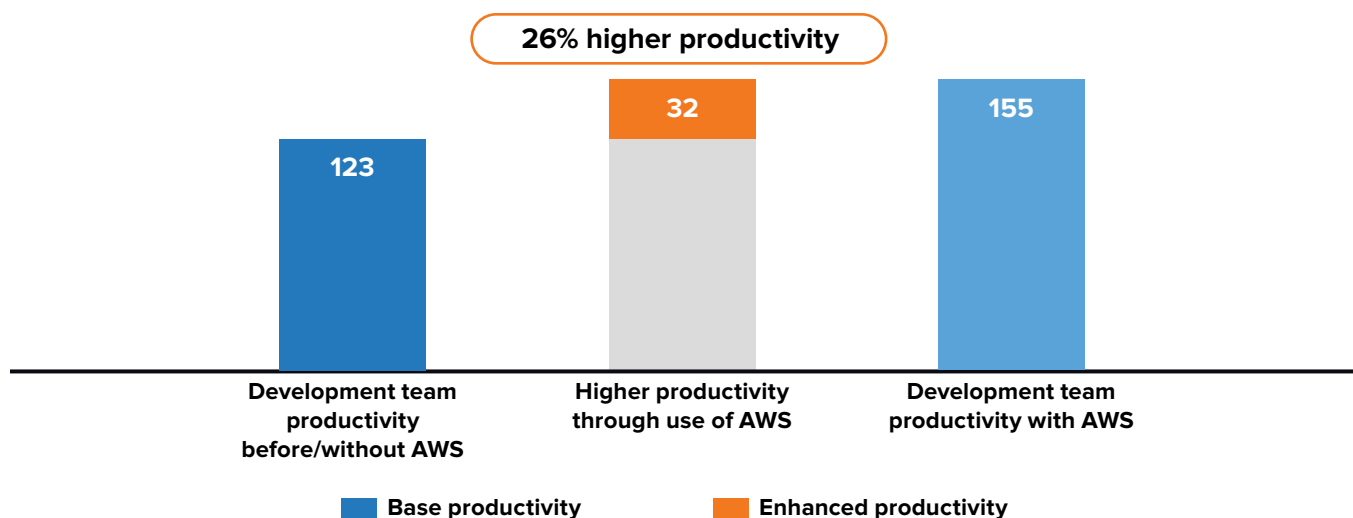
► **Substantial change in IT agility to support development (head of cloud, telecommunications, Australia):**

*"We're 100% more agile with AWS. The development team can have new compute resources within 30 minutes. On premises that would take, from a request going in, anywhere from two to four weeks."*

Development teams that can more readily meet user and business demand for new functionality provide more value to their organizations. IDC calculates that interviewed organizations' development teams achieved an average 26% improvement in productivity with AWS. This gain in productivity for development teams reflects their ability to better serve and actively support business initiatives and operations (see **Figure 8**).

**FIGURE 8**  
**Development Team Productivity Gains**

(Equivalent productivity — FTEs per organization)



n = 15; Source: IDC in-depth interviews, February 2022

## Minimizing Operational Risk and Related Costs

Study participants from the APAC region reported significant performance benefits with AWS. They cited not only improved performance at an application level but also much-enhanced reliability of applications and services running on AWS. And for many of the study's participants, AWS replaced on-premises environments comprising aging infrastructure prone to business-impacting outages that required significant staff time and attention to update and patch.

Interviewed APAC-based organizations cited key performance-related benefits of AWS, including architecting more secure and robust IT platforms, resolving problems faster, and provisioning resources to address performance issues more readily. These benefits drove improved ability to avoid impactful unplanned outages and thus reduce operational risk related to performance.

### Study participants provided specific examples of their experiences with AWS, including:

- ▶ **Direct support to ensure effective SecOps (data scientist, telco, Philippines):**  
*"AWS helps us with our security architecture by confirming that our team is actually able to use that component properly, and they provide guidance on how to better secure some of the application-specific components to make sure we have proper knowledge to implement things."*
- ▶ **Much faster to resolve potential problems (data science lead, insurance, Hong Kong):**  
*"The most obvious change with AWS is the IT time to find the solutions and time to implement it. If something is wrong, it's a lot quicker to fix it — 2–3 times faster — and it requires so much less staff to fix problems."*
- ▶ **More flexible access to capacity to provide reliability and scalability (enterprise systems architect, professional services, Japan):**  
*"We changed how our overall IT system looks with AWS; we added more servers than we had before, but smaller ones with the possibility to scale up so we can save some money. We also increased performance by about 50% because we built bigger clusters on the AWS side."*

**Table 4** (next page) shows that study participants' move to AWS has resulted in significant reductions in unplanned downtime and associated productivity and business losses. Ultimately, interviewed APAC organizations reported losing 72% less productive time and an impressive 89% less revenue due to unplanned outages. These improvements reflect not only a minimization of costs associated with such outages but also much-reduced operational risk that comes along with outages that affect not only employees but customers and even brand perception and company goodwill.

TABLE 4

## Impact on Unplanned Downtime KPIs

|                                                           | Before/<br>Without AWS | With AWS  | Difference | Benefit |
|-----------------------------------------------------------|------------------------|-----------|------------|---------|
| Number of unplanned outages per year                      | 16                     | 8         | 8          | 49%     |
| Mean time to repair (hours)                               | 5                      | 1         | 4          | 78%     |
| Hours of productive time lost per user per year           | 4                      | 1         | 3          | 72%     |
| Productivity loss per year in FTEs per organization       | 5                      | 1         | 4          | 72%     |
| Value of lost productivity time per organization per year | \$366,100              | \$101,700 | \$264,400  | 72%     |
| Value of lost revenue per year per organization           | \$2.7M                 | \$0.3M    | \$2.4M     | 89%     |

n = 14; Source: IDC in-depth interviews, September 2021

## Realizing Better Business Results

APAC organizations participating in this study reported a broad range of business enablement benefits related to AWS use, including improved performance, enhanced agility and scalability, and improved user/customer experience. These benefits coalesced to enable study participants to better serve existing customers, address new opportunities, and generate higher revenue.

Interviewed APAC organizations attributed much of their ability to generate increased revenue to the extent to which AWS has enabled them to adopt and leverage new business-benefiting technologies, including artificial intelligence and machine learning capabilities. In addition, they noted that their AWS environments promoted a degree of nimbleness in adapting to, and leveraging, fluctuating market conditions. For example, although IDC classified development team productivity gains as IT staff–related benefits for this study, enhanced development capabilities often lead directly to improved ability to address and win business opportunities.

**Study participants provided specific examples of how AWS has improved their ability to establish and maintain competitive differentiation in their markets:**

► **Ensures the ability to compete on new functionalities and trends (head of IT infrastructure, financial services, Singapore):**

*“Our market is changing quite fast, so we have to be fast to adapt to it, and AWS helps us back into the changing environment. For example, competitors may have some promotion, and we don’t want to be left behind, so we want to have a similar promotion. Same with a new function, we want to have new functions to compete, so we need to change and react to our competitors’ new initiatives.”*

▶ **ML-focused customer recommendations to enable specific service (data scientist, telco, Philippines):**

*“We have machine learning through AWS that recommends the next product that the customer should buy or try. We also introduced a ‘loan business’ where the subscriber can actually borrow a small amount to continue using the services for emergency calls or to subscribe to a data promo, and we’ve doubled our revenue there with AWS.”*

▶ **Redundancy required for business-critical applications (enterprise systems architect, professional services, Japan):**

*“One of our fundamental goals of moving to AWS was to reduce downtime, especially on our external websites because, if the website goes down, the customers can’t use it. ... Our recruitment application we’ve built into the AWS environment is servicing a business of approximately 800 recruiters, and without it, we can’t do anything. Our revenue is about \$1 million a day, so one day out is very expensive and why we have massive amounts of resiliency in that application with AWS.”*

▶ **Scale and new technologies (head of cloud, telecommunications, Australia):**

*“One of the biggest things for us with AWS is getting scale when we need it ... but the biggest benefit is all of the services that come with Amazon, including AI/ML and the storage — if we were on premises, it would be slower or we wouldn’t be able to do all of this, and it helps us increase revenue.”*

Interviewed APAC organizations consistently linked the use of AWS to direct business gains. They described moving faster to meet changing customer demand patterns, scaling their businesses up and down on little notice, and providing improved user and customer experiences that encourage customer retention and loyalty.

## They spoke in specifics about how AWS has enabled their businesses:

▶ **Faster onboarding and the ability to leverage AWS support to meet customer requirements (enterprise systems architect, professional services, Japan):**

*“With AWS, we can onboard a new customer in about half the time as with our on-premises infrastructure. We no longer have problems with the aging hardware, which is one of the main reasons we moved to AWS. Some customers require additional protection and guarantees for all of their data, and we could not provide that because there was no more support from the vendor on our old hardware.”*

▶ **Faster to meet business demand (head of IT infrastructure, financial services, Singapore):**

*“With AWS, we can fulfill business demands in a much shorter time now, at least 20–30%. ... If we have a campaign or something that needs infrastructure to support the events, it only takes a week now to prepare for it. Before we had to create outcomes, build the OS, do network patching, and allocate IP addresses. Now, with AWS, everything is there, so we just turn it on and use it.”*

▶ **Ensuring quality of user experience during demand surge (vice president of technology, media and entertainment, India):**

*“When the pandemic hit in March of 2020, we had a huge spike in demand for videos for binge watching over different platforms, whether it was mobile, tablet, television, or laptop. Our normal peak load of 60–70% for VOD went to 165–180%. Because we were on AWS, we were able to manage that with proper efficiency as well as user experience.”*

IDC calculates that study participants will realize a significant average boost to revenue of \$22.0 million per year, or \$45,600 per AWS EC2 virtual instance. For purposes of IDC's financial model, a 15% operating margin assumption is applied to calculate net revenue/operating income, which results in average net revenue gains of \$3.3 million per year per organization, or \$6,800 per AWS EC2 virtual instance (see **Table 5**).

**TABLE 5****Business Productivity Benefits — Higher Revenue**

|                                        | Per Organization | Per EC2 Virtual Instance |
|----------------------------------------|------------------|--------------------------|
| <b>Revenue impact</b>                  |                  |                          |
| Total additional revenue per year      | \$22.0M          | \$45,600                 |
| Assumed operating margin               | 15%              | 15%                      |
| Total additional net revenue per year* | \$3.3M           | \$6,800                  |

n = 15; Source: IDC in-depth interviews, February 2022

\* IDC model assumes a 15% operating margin to calculate net revenue gains.

## ROI Summary

**Table 6** (next page) presents IDC's analysis of the financial benefits and costs related to the use of Amazon Web Services by interviewed APAC organizations. IDC calculates that on a per-organization basis, interviewed organizations will achieve total five-year discounted benefits of \$35.6 million, or \$73,600 per AWS EC2 virtual instance, based on enhanced business benefits, improved IT staff efficiencies, faster and more efficient software development, reductions in unplanned downtime, and optimized IT infrastructure-related costs. These benefits compare with projected total discounted investment costs of \$8.5 million per organization over five years. IDC calculates that at these levels of benefits and investment costs, interviewed organizations will achieve a five-year ROI of 318% and break even on their investment in AWS in 12 months.

**TABLE 6**  
**ROI Analysis**

|                                   | Five-Year Average per Organization | Five-Year Average per AWS EC2 Virtual Instance |
|-----------------------------------|------------------------------------|------------------------------------------------|
| <b>Benefit (discounted)</b>       | \$35.6M                            | \$73,600                                       |
| <b>Investment (discounted)</b>    | \$8.5M                             | \$17,600                                       |
| <b>Net present value (NPV)</b>    | \$27.1M                            | \$56,000                                       |
| <b>Return on investment (ROI)</b> | 318%                               | 318%                                           |
| <b>Payback period</b>             | 12 months                          | 12 months                                      |
| <b>Discount rate</b>              | 12%                                | 12%                                            |

n = 15; Source: IDC in-depth interviews, February 2022

## Challenges/Opportunities

**In pursuing digital innovation and making decisions regarding cloud adoption, IDC sees organizations successfully addressing the following challenges:**

- ▶ Speed of provisioning infrastructure and moving from an IT ticket–request type of process to self-service, thereby empowering development teams to move quickly
- ▶ Ability to scale (and speed of scaling) the infrastructure up and down with containerization and serverless, thereby reducing overhead costs significantly
- ▶ Team size efficiencies for automated patching and updates, enabling the need for fewer people even as infrastructure capacity grows
- ▶ Minimizing costs and staff time related to unplanned outages and security issues, allowing their time to be more productively engaged
- ▶ Enabling innovation with machine learning and artificial intelligence to create competitive differentiation, deliver cost savings, and increase revenue
- ▶ Empowering developers to deliver new applications and functionality to market rapidly (e.g., within three months instead of six or in days/weeks instead of months)
- ▶ Needing flawless operation and getting attention from executives, not because of outages, but for enabling the business

Successfully addressing these challenges represents direct lines of questioning for cloud providers as you align technology and architecture decisions with the overarching digital strategy of your organization.

# Conclusion

Cloud adoption continues at a rapid pace, with IDC forecasting a CAGR of 29% through 2025 in the combined IaaS and PaaS markets. Organizations that adopt cloud technologies find that the combination of lower-cost cloud-delivered services and agility from open source and cloud-native application development practices leads to competitive advantage because technology innovation and competitiveness are tightly linked. Today's C-suite increasingly sees portfolios of cloud-delivered services covering analytics, business applications, blockchain, security/identity/compliance, storage, database, development tools, compute, media services, hybrid architecture, Internet of Things, and machine learning as critical for delivering their own differentiated services to customers.

This IDC study shows how organizations in diverse and fast-changing APAC markets have leveraged various AWS solutions to operate and compete more efficiently and effectively. Interviewed AWS customers in APAC reported optimizing their costs to run equivalent workloads and have translated staff efficiencies to focus on business enablement and innovation. On the business operational side, study participants cited improved agility, performance, and resiliency as driving significant value in increased revenue, reduced risk, and operational efficiencies. Overall, IDC projects that interviewed APAC customers will realize more than \$4 of benefits for every \$1 invested in AWS (five-year ROI of 318%), reflecting the strong value proposition of AWS for them.

## Appendix A:

### Methodology

IDC's standard Business Value/ROI methodology was utilized for this project. This methodology is based on gathering data from organizations currently using Amazon Web Services solutions as the foundation for the model.

**Based on interviews with organizations using AWS, IDC performed a three-step process to calculate the ROI and payback period:**

- 1. Gathered quantitative benefit information during the interviews using a before-and-after assessment of the impact of using Amazon Web Services.** In this study, the benefits included IT infrastructure cost savings, IT staff and development team efficiencies and productivity gains, reduced costs associated with risk, and higher revenue.
- 2. Created a complete investment (five-year total cost analysis) profile based on the interviews.** Investments go beyond the initial and annual costs of using Amazon Web Services and can include additional costs related to migrations, planning, consulting, and staff or user training.
- 3. Calculated the ROI and payback period.** IDC conducted a depreciated cash flow analysis of the benefits and investments for the organizations' use of Amazon Web Services over a five-year period. ROI is the ratio of the net present value (NPV) and the discounted investment. The payback period is the point at which cumulative benefits equal the initial investment.

**IDC bases the payback period and ROI calculations on a number of assumptions, which are summarized as follows:**

- ▶ Time values are multiplied by burdened salary (salary + 28% for benefits and overhead) to quantify efficiency and manager productivity savings. For purposes of this analysis, based on the geographic locations of the interviewed organizations, IDC has used assumptions of an average fully loaded salary of \$100,000 per year for IT staff members and an average fully loaded salary of \$70,000 per year for non-IT staff members. IDC assumes that employees work 1,880 hours per year (47 weeks x 40 hours).
- ▶ The net present value of the five-year benefits is calculated by subtracting the amount that would have been realized by investing the original sum in an instrument yielding a 12% return to allow for the missed opportunity cost. This accounts for both the assumed cost of money and the assumed rate of return.
- ▶ Because IT solutions require a deployment period, the full benefits of the solution are not available during deployment. To capture this reality, IDC prorates the benefits on a monthly basis and then subtracts the deployment time from the first-year savings.

*Note: All numbers in this document may be inexact due to rounding.*

## Appendix B:

### Quantified Benefits of Use of AWS

**Table 7** (next page) provides details about the financial benefits that IDC calculates interviewed APAC organizations will achieve.

TABLE 7

## Annual Quantified Financial Benefits

| Category of Value                                      | Average Quantitative Benefit                                              | 15% Margin Assumption Applied | Calculated Average Annual Value* |
|--------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------|----------------------------------|
| IT infrastructure cost savings                         | 25% lower average cost, with savings of \$615,300 per year                | No                            | \$521,600                        |
| IT infrastructure and administrative team efficiencies | 38% efficiency, saving 32 FTEs                                            | No                            | \$2.72 million                   |
| IT security team efficiencies                          | 29% efficiency, saving 5 FTEs                                             | No                            | \$413,100                        |
| Help desk team efficiencies                            | 34% efficiency, saving 3 FTEs                                             | No                            | \$255,000                        |
| DBA team efficiencies                                  | 22% efficiency, saving 4 FTEs                                             | No                            | \$296,300                        |
| Application development team productivity gains        | 26% higher productivity, worth 32.1 additional FTEs                       | No                            | \$2.7M                           |
| Productivity gains with reduced unplanned downtime     | 3.1 hours saved per user, 4 FTEs of productivity gained (\$70,000 salary) | No                            | \$224,200                        |
| Revenue gains with reduced unplanned downtime          | 89% less revenue lost, saving \$2.4 million per year per organization     | Yes                           | \$308,200                        |
| Compliance team productivity gains and cost savings    | 33% higher productivity, equal to gain of 0.2 FTEs                        | No                            | \$18,800                         |
| Revenue gains with business enablement                 | \$22.0 million in revenue gained per year                                 | Yes                           | \$2.8M                           |
| Analytics team productivity gains                      | 3% higher productivity, equal to gain of 0.8 FTEs                         | No                            | 44,500                           |
| <b>Total annual benefits using AWS</b>                 |                                                                           |                               | <b>\$10.3M</b>                   |

n = 15; Source: IDC in-depth interviews, February 2022

(\* Includes 9.1 months deployment time in year 1)

# Appendix C:

## Additional Quotes

Selected additional quotes from interviews with APAC organizations using AWS are as follows:

► **Opportunity to leverage new approaches like containerization, enterprise systems architect, professional services, Japan:**

*"The next big step for us with AWS is to look at how we containerize all of our applications because we haven't done that very much yet. We've actually got a major rebuild coming up, and we're looking toward how we containerize all of the applications, building with microservices, and to what degree that entire layer could be managed."*

► **AWS providing platform necessary for timely and effective acquisitions/expansion, regional head of APAC, manufacturing, Singapore:**

*"We want more presence and diversification in [our region] and we are growing by acquisitions. ... We've acquired significant manufacturing operations that we bring under our hold and establish our governance using our R&D in order to expand our business. Acquisitions are aided by going into a cloud platform, and AWS is something on which we can quickly expand our IT platform because we want to grow fast. With the pandemic over the past two years, getting on-premises hardware on time has been a continuous challenge. We don't have that problem with AWS."*

► **Scalability to meet customer demand and avoiding problems associated with provisioning on-premises equipment, enterprise systems architect, professional services, Japan:**

*"We chose AWS for two main reasons. First, our new customers expect us to be able to scale up quickly. Second was that we saw delays for hardware that is in our existing data project due to COVID-19. This was the turning point that we started using more and more solutions on AWS."*

► **Ease of scalability that benefits IT teams, vice president of technology, media and entertainment, India:**

*"The ability to scale up the infrastructure with AWS is the biggest advantage. The single control is very helpful for managing the load to shut down any instance for cost savings, and the consumption model helps because we can do a small project, testing, or some kind of lab work very fast. The foundation is already there, you just build up what you want, and that adds to a lot of convenience and operational efficiency."*

► **Peak workload savings, vice president of technology, media and entertainment, India:**

*"We did an evaluation and calculation of the on-premises solution versus AWS, and we are saving around 40% during the peak workloads with AWS. Another key benefit is the staff efficiency because we have a limited staff and AWS manages the overall program governance as well as the instance part."*

# About the Analysts

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Dr. Lara Greden is research director for IDC's Platform as a Service (PaaS) practice. Greden's research focuses on platforms for application development on private, public, and hybrid clouds and on edge deployments. She directs research into the competitive markets of cloud platforms and application development and deployment services that are enabling digital transformation, including integration, containers, serverless computing, big data, AI, ML, predictive analytics, IoT, and other emerging technologies.

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Matthew is responsible for carrying out custom business value research engagements and consulting projects for clients in a number of technology areas with a focus on determining the return on investment (ROI) of their use of enterprise technologies. Matthew's research often analyzes how organizations are leveraging investment in digital technology solutions and initiatives to create value through efficiencies and business enablement.

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